DP Computer Science: A 2.1.1 Network Characteristics

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Define and explain the **purpose** of computer networks
- Identify and describe the characteristics of different network types: **LAN, WAN, PAN, VPN**
- Explain the importance of **network security** and identify common security threats
- Apply their understanding of networking concepts to **real-world scenarios**

Key Vocabulary

Term	Definition
Network	A collection of devices connected together to share resources and communicate
LAN (Local Area Network)	A network that covers a small geographical area, such as a home, school, or office
WAN (Wide Area Network)	A network that covers a large geographical area, often connecting multiple LANs
PAN (Personal Area Network)	A network for personal devices, typically within a range of a few meters
VPN (Virtual Private Network)	A secure network connection that encrypts data and masks IP addresses
Bandwidth	The maximum data transfer rate of a network (measured in bits per second)
Latency	The delay between sending and receiving data (measured in milliseconds)
Topology	The physical or logical arrangement of devices in a network
Node	Any device connected to a network (computer, printer, switch, router, etc.)

IB ATL Skills

Skill	Description
Communication	Communicating information and ideas effectively using appropriate terminology
Collaboration	Working collaboratively with peers to complete activities and solve problems

Skill	Description
Thinking	Applying thinking skills critically and creatively to recognise and approach complex problems

2. What is a Network?

Purpose of Networks

Networks connect devices for:

Purpose	Description
Communication	Sharing messages, emails, video calls, and instant messaging
Resource Sharing	Sharing hardware (printers, scanners), software, and files
Information Access	Accessing the internet, cloud services, and shared databases
Collaboration	Working together on shared documents and projects

Network Diagram

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3. Types of Networks

Network Type	Description	Range	Examples
PAN (Personal Area Network)	Connects personal devices within a very small area	Up to ~10m	Bluetooth headset, smartwatch pairing, file transfer between phone and laptop
LAN (Local Area Network)	Connects devices in a small geographical area	Building or campus	School network, office network, home Wi-F

Network Type	Description	Range	Examples
WAN (Wide Area Network)	Connects devices across large geographical areas	City, country, global	The internet, multinational company netwo
VPN (Virtual Private Network)	Creates a secure, encrypted connection over the internet	Anywhere with internet	Secure remote work, bypassing geo-restric

Comparison Table

Feature	PAN	LAN	WAN	VPN
Size	Smallest	Small	Largest	Variable
Ownership	Personal	Organisation	Multiple organisations	Service provider
Speed	Low/Medium	High	Variable	Limited by internet
Security	Low	High	Low (public)	High (encrypted)
Cost	Low	Medium	High	Variable
Examples	Bluetooth, USB	Ethernet, Wi-Fi	Internet	ExpressVPN, NordVPN

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4. Network Characteristics

Bandwidth

Aspect	Description
Definition	The maximum amount of data that can be transmitted over a network in a given time
Measurement	Bits per second (bps), kilobits (kbps), megabits (Mbps), gigabits (Gbps)
Higher Bandwidth	Faster data transfer → better performance for large files, streaming
Limited Bandwidth	Slower transfer → buffering, longer download times

Analogy: Bandwidth is like the width of a pipe—wider pipe = more water (data) can flow through.

Latency

Aspect	Description
Definition	The delay between sending and receiving data over a network
Measurement	Milliseconds (ms)
Low Latency	Fast response → good for gaming, video calls, real-time applications
High Latency	Slow response → delays, lag, poor experience

Analogy: Latency is like the time it takes for a message to travel—speed of delivery, not the amount delivered.

Bandwidth vs. Latency:

Comparison	Bandwidth	Latency
Definition	Data transfer capacity	Data transfer delay
Affects	How much data can be sent	How quickly data arrives
Critical For	Large file transfers, streaming	Gaming, video calls
Analogy	Width of a pipe	Time for water to travel through pipe

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Topology

Type	Description	Advantages	Disadvantages
Bus	All devices connected to a single cable	Simple, cheap	Single point of failure
Star	All devices connected to a central device (switch/hub)	Fault-tolerant; easy to add devices	Central device is a single point of failure
Ring	Devices connected in a circular loop	Predictable transmission	Single break disrupts whole network
Mesh	Every device connected to every other	Highly redundant; fault-tolerant	Expensive; complex

Visual Summary:

	Bus	Star	Ring	Mesh

Diagram	`---●---●---●---`	`●───┐`
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5. Network Security

Common Threats

Threat	Description	Example
Unauthorised Access	Gaining access to a network without permission	Hacking, password cracking
Malware	Malicious software that damages or disrupts systems	Viruses, ransomware, spyware
Phishing	Tricking users into revealing sensitive information	Fake emails, fake websites
Denial of Service (DoS)	Overwhelming a network to make it unavailable	DDoS attacks flooding servers
Man-in-the-Middle	Intercepting communication between two parties	Intercepting Wi-Fi traffic

Security Measures

Measure	Description
Encryption	Scrambling data to prevent unauthorised access
Firewall	Software/hardware that monitors and controls network traffic
Authentication	Verifying user identity (passwords, 2FA, biometrics)
Antivirus/Antimalware	Software that detects and removes malicious software
Network Monitoring	Continuous monitoring for suspicious activity

VPN (Virtual Private Network)

Aspect	Description
What it does	Creates a secure, encrypted connection over the internet
Key Features	Encrypts data, masks IP address, hides online activity
When to use	Remote work, public Wi-Fi, bypassing geo-restrictions
Examples	ExpressVPN, NordVPN, ProtonVPN

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6. Lesson Sequence

Lesson Plan

Phase	Time	Activity
Engage	10 min	"What devices in your home are connected to a network?" Brainstorming; discussion of what connects to the internet
Explore	15 min	Introduce network purpose and types (LAN, WAN, PAN, VPN); use visuals and real-world examples
Explain	15 min	Key characteristics: bandwidth, latency, topology; compare and contrast with analogies
Elaborate	15 min	Group scenario activity: design a network for a given scenario (e.g., school, small business, gaming house)
Evaluate	5 min	Exit ticket: key questions on network types and characteristics

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Scenario-Based Activity

Task: In groups, design a network for a given scenario.

Scenario	Requirements
School Network	Connect 30 computers, printers, and internet access; must be secure

Scenario	Requirements
Small Business	Connect 10 workstations, file server, printer, and allow remote access for employees
Gaming House	Connect multiple gaming PCs, consoles, and smart TVs; low latency is critical
Home Setup	Connect two laptops, a smart TV, and a printer; set up wireless access

Questions for Each Scenario:

1. What network type(s) would you use?
2. What security measures would you implement?
3. What hardware would you need?

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7. Assessment Activities

Assessment Type	Description
Class Participation	Assess engagement and contributions during discussions and activities
Interactive Quiz	Use online tools (Quizlet) to assess understanding of network types and characteristics
Scenario-Based Activity	Assess application of networking concepts to real-world situations
Exit Ticket	Gauge understanding and identify areas needing further clarification

Exit Ticket Questions:

1. What is the purpose of a computer network?
2. Explain the difference between a LAN and a WAN.
3. What is a VPN, and why might someone use one?
4. Give two examples of network security threats.

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8. Differentiation

Level	Strategy
Support	Provide visual aids, concise definitions, and real-world examples
Challenge	Encourage advanced students to explore complex scenarios or research emerging network technologies (e.g., 5G, SDN, IoT)

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9. Resources

- **Presentation** (Slide Deck)
- **eBook** (A2.1.1 with quiz)
- **Worksheet:** Network types comparison + scenario questions
- **Quizlet** (A2.1.1 vocabulary flashcards)

10. Summary: Key Takeaways

Concept	Key Point
Network Purpose	Communication, resource sharing, information access
LAN vs WAN	LAN = small area (building); WAN = large area (country/global)
PAN	Personal devices, short range (Bluetooth)
VPN	Secure, encrypted connection over the internet
Bandwidth	Data capacity — "width of the pipe"
Latency	Data delay — "speed of delivery"
Topology	Physical arrangement of devices (star, bus, ring, mesh)
Security	Encryption, firewalls, authentication, VPNs

> "*Understanding networks is essential for navigating our interconnected world—from sending an email to streaming a movie, networks are everywhere.*"